Serena Noboudem

nobserena@gmail.com | 4045999575 | Boston, MA | LinkedIn | GitHub | Portfolio

EDUCATION

Boston University, Masters in Computer Science — Sep 2025 – Jan 2027 | Boston, MA

- **Coursework**, CS630: Graduate Algorithms, CS655: Computer Networks, CS651: Distributed Systems, CS541: Applied Machine Learning, CS598 G1: AI, Cyber, & Entrepr, CS955: Independent Research (AI Security), CS598 A1: Agentic AI

Kennesaw State University, Bachelor of Science in Computer Science — May 2025 | Kennesaw, GA

SKILLS

- **Languages&Databases:** Java, Spring Boot, Python, Go, JavaScript/TypeScript, React, MySQL, REST, gRPC, Elasticsearch, Redis
- **Infra&Tools:** API Gateways, Caching, Concurrent programming, Git/GitHub, CI/CD (Jenkins), Docker
- **Security&Networking:** Authentication/Authorization (OAuth2, RBAC, JWT), Network protocols (DNS, DHCP, TCP/IP)

WORK EXPERIENCE

Software Engineer, Intern, Netflix — May 2026 – Aug 2026

- Designed and implemented a **safety-critical failure injection architecture** for Netflix's internal API gateway, enabling 2+ engineer teams to safely **simulate dependency outages** on a scoped slice of production traffic
- Built a full-stack continuous **health-probing system** end-to-end in **Spring Boot, React,** and **gRPC** with a shared **protobuf contract**, enabling **cross-team engineers** to analyze and visualize synthetic traffic sessions against production gateways on a unified dashboard measuring **100% traffic latency, status,** and **proxy health signals.**
- Engineered the **concurrent request-dispatch** and **scheduling core** of the system using Java **virtual threads** and in-memory connection pooling to fire **~150req/s** of synthetic HTTP traffic
- Designed the **reconciliation-loop architecture** that reduced data store reads and latency by **33.3%**

Graduate Researcher, GRS, BU — Jan 2026 – May 2026

Advised faculty supervision | *Report*

- Designed an **autonomous evolutionary pipeline** that created and optimized visual jailbreak attacks against multimodal LLMs, outperforming hand-crafted baselines with a **self-improving system**.
- Applied MAP-Elites strategy to drive in-context optimization across **90+ unattended rounds** and thousands of model inferences.
- Achieved **87.5%** training and **90%** validation attack success rate against a multimodal LLM (Qwen2-VL-7B).

Research Assistant, KSU - College of Computing — Oct 2024 – May 2025

Abstract | *Poster*

- Crafted & executed adversarial attacks (MIFGSM, FGSM) on deep learning models using the MNIST dataset, executing **40+ attack iterations,** applying **threat modeling techniques** to assess model robustness and uncover key vulnerabilities
- Conducted **cross-model evaluation** across **4+ architectures,** analyzing **5,000+ misclassified samples,** visualizing error patterns, and improved **model robustness by 80%**
- Delivered research findings at the **Annual KSU Symposium**, effectively translating complex technical concepts into accessible insights for an audience of **50+** faculty, researchers, and peers.

Software Engineer Intern, Afriland First Bank — Jun 2023 – Aug 2023

- Performed **threat modeling** using the **STRIDE methodology** to identify and prioritize system vulnerabilities, resulting in the mitigation of **15+** high-priority security risks before deployment.
- Engineered a **secure authentication system** using **OAuth2, JWT,** and **2FA** in Java for a banking app, achieving **100% compliance** with banking regulations and reducing unauthorized access attempts by **70%**.

PROJECTS

Task Mesh (Python | Redis | Pydantic | Gemini)

- Architected a distributed task mesh, effectively mitigating state-syncing errors and systemic failures, by utilizing **Redis streams** for queuing and a custom **Dead-Letter Queue recovery agent** to process and recover dead tasks.

FinHub (Spring boot | React.js | MySQL)

- Developed a full-stack finance tracker web app in **Spring Boot** and **React.js** to retrieve and analyze financial data from **2+** bank accounts, to generate personalized spending insights and budget reports
- Integrated the Plaid API via **WebClient** with secure authentication, enhancing HTTP request handling efficiency by **10%** and ensuring safe, scalable access to user data with reduced latency

DNS Server (Python)

- Engineered a multithreaded DNS server in Python from scratch using **Socket programming,** handling **500+ QPS**, achieving **99%** resolution accuracy through efficient request parsing and high-throughput